



Tailoring the properties of carbon molecular sieves membranes for the separation of propionic acid from aqueous solutions

Arash Rahimalimamaghani^{a,c}, David A. Pacheco Tanaka^{a,b}, Margot A. Llosa Tanco^{a,b}, Fausto Gallucci^{a,c,*}

^a Inorganic Membranes and Membrane Reactors, Sustainable Process Engineering, Chemical Engineering and Chemistry, Eindhoven University of Technology, De Rondom 70, 5612 AP, Eindhoven, the Netherlands

^b TECNALIA, Basque Research and Technology Alliance (BRTA), Mikeletegi Pasealekua 2, 20009, Donostia-San Sebastian, Gipuzkoa, Spain

^c X:MEM BV, Groene Loper 3, 5612 AE, Eindhoven, the Netherlands

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ABSTRACT

In the fermentative production of propionic acid (PA), the major problem with batch fermentation systems is the strong inhibitory effect of PA on the production yield; one way to increase the yield is the in-situ removal of PA by using pervaporation. Acetic acid (AA) is the most important by-product in the fermentation; therefore, the membrane should be able to remove selectively PA from an aqueous solution containing AA. Considering that PA is more hydrophobic than AA and their kinetic diameter are 0.480 and 0.436 nm respectively, hydrophobic membranes with main pores in the range of around 0.5–0.6 nm with high permeation are required.

Supported thin Carbon Molecular Sieve Membranes (CMSM) were prepared by the dip coating a porous alumina support into a solution containing resorcinol phenolic resin as carbon source. The hydrophobicity was obtained by carbonizing the polymer at temperatures higher than 750 °C and adding polyvinyl butyral (PVB) as pore forming agent and carbon contributor. PA with 88 % of purity was obtained by pervaporation of an aqueous solution containing 5 % of PA and 5 % of AA using a CMSM carbonized at 850 °C containing 1 % of PVB in the dipping solution.

1. Introduction

Propionic acid, a three carbon carboxylic acid, is intensively used in the food and animal feed industry as anti-bacterial agent in water, and in the production of herbicides, pharmaceuticals, polymers and perfumes [1]. The PA market size is expected to reach 412 kilotons in 2024 and 485 kilotons by 2029, Europe accounts for the largest market share and the Asia-Pacific region is estimated to have the highest grow in the next years [2]. The production of PA by fermentation has been studied for more than 450 years [3], however, the petrochemical production still is more economical [4]. The production of PA from industrial wastes such as glycerol, wood pulp waste liquors, lignocelluloses, can make biomass-based PA economically more competitive [5–7]. The major problem with a batch system of fermentation is the strong inhibitory effect of PA on the production yield, especially when the pH is below 6 [8,9].

Acetic acid is the most important by-product produced during

propionic fermentation and it can impair PA production (due to its toxicity) and recovery in downstream processing [4]. Low acid concentration ensures higher product yield and lower amounts of by-products; one way to increase the yield is to remove PA from the fermentation process, this process is known as extractive fermentation or in situ recovery [10]. Several techniques have been reported to remove PA such as solvent extraction, electrodialysis and resin adsorption [4]. One effective separation method is pervaporation (PV); in this process, a membrane is in contact with the fermentation liquid, and in one side of the membrane vacuum is applied to allow the permeation of gas stream through the membrane. Since in PV the fermentation liquid is not subject to harsh condition, this technique has little impact in the microorganisms, fermentation nutrients and substrates [10]. In PA pervaporation separation process, key parameters such as high PA permeance, PA perm-selectivity and stable performance require a tailor-made membrane regarding hydrophobicity, pore size distribution (PSD) and porosity. In polymeric membranes, enhancing both PA

* Corresponding author. Inorganic Membranes and Membrane Reactors, Sustainable Process Engineering, Chemical Engineering and Chemistry, Eindhoven University of Technology, De Rondom 70, 5612 AP, Eindhoven, the Netherlands.

E-mail address: f.gallucci@tue.nl (F. Gallucci).

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perm-selectivity and permeance faces challenges due to the existing problems such as stability of membrane, fouling and tradeoff between perm-selectivity and permeance which hinders from industrial application [11].

Carbon Molecular sieve membranes (CMSMs) are porous (0.25–0.9 nm pore size) membranes, as a product of the carbonization of thermosetting polymers such as polyimides and phenolic resins [12–14]. The pore size, pore size distribution hydrophobicity/hydrophilicity can be tailored by changing the polymer, carbonization conditions, the addition of nano additives in their structure and post treatments [15,16]. Gas permeation through CMSM is the combination of molecular sieving (molecules smaller than the pore size will permeate) and adsorption diffusion (depending on the interaction of the molecules with the pores). Selective adsorption favors the permeation and at the same time blocks the pores to the permeation of other less selective molecules. For effective separation of PA from low concentration aqueous phase in membrane bio-reactors, the CMSM should be hydrophobic, contain narrow pore size distribution (PSD) and should be stable chemically/physically for long term performance.

In this work, for the first time, CMSM will be used for the separation of PA from an aqueous solution containing acetic acid. Since PA is more hydrophobic than AA, supported hydrophobic CMSMs were prepared by dipping a porous support in a solution containing a resorcinol-formaldehyde resin with various concentrations of polyvinyl butyral (PVB), as sacrificial pore former and hydrophobic enhanced agent, and then, carbonized at 750 or 850 °C. The pore size distribution of the membranes (determined by He perm-porometry) and water adsorption studies show that the pore size distribution and hydrophilicity changes with the carbonization temperature and addition of PVB.

2. Experimental

2.1. Synthesis: materials and methods

Formaldehyde (37 %), propionic acid (PA) (99 %) and acetic acid (AA) (99–100 %) from VWR, KOH pellets, *N*-methyl-2-pyrrolidone (NMP, 99.5 %), polyvinyl butyral (PVB) (Mw 50–80 K) and resorcinol (99 %) were purchased from Merck (Sigma-Aldrich) and used without further purification. Asymmetric tubular supports with ZrO₂ on the outer side with an average pore size of 120 nm, outer diameter of 10 mm and inner diameter of 7 mm were supplied by Inopore GMBH.

The CMSMs were prepared using resorcinol-formaldehyde resin, as carbon source, by the dip-dry (polymerization)- carbonization method reported before [15] (Fig. S1). In short, first, a resorcinol-formaldehyde oligomer was prepared by the condensation reaction of resorcinol and formaldehyde in water at 90 °C using KOH as catalyst for 3 h. The product was separated by centrifugation, washed with water, and dried under vacuum at 40 °C for 24 h. The dipping solution was prepared by dissolving 16 g of the oligomer in 80 g of NMP, then 2.8 g of formaldehyde solution and 0.1 g of KOH were added to the solution, and the polymerization was carried out for 30 min. Then PVB was added to the solution to obtain concentrations of 0, 0.5 and 1 wt%: of PVB (P0, P0.5 and P1 respectively) via high shear mixing for 30 min with 2000 rpm for two cycles (Thinky 250). The dipping solution was deposited on the porous support by dip coating. The coated supports were then carbonized under vacuum (14 mbar) at 750 or 850 °C; the carbonization protocol is shown in Fig. S2. The membranes were named with the temperature of carbonization followed by the concentration of PVB in the dipping solution; thus, 750P0 and 850P1 correspond to the dipping solutions of the membranes carbonized at 750 °C without the addition of PVB and 850 °C with 1 % of PVB respectively.

Non-supported CMSM films for physicochemical characterization were obtained from the remaining dipping solution by pouring in Teflon evaporating dishes and heating at 100 °C for 24 h. The obtained films were carbonized at the same condition of the supported CMSM.

2.2. Characterization of CMSM

Hydrophobicity/Hydrophilicity analysis. Non-supported CMSMs samples were powdered and sieved to 20–40 µm particle size, weighted and placed inside a 100 Relative Humidity (RH) chamber for 7 days. After saturation with water vapor, the samples were weighted again, and the amount of water adsorbed calculated.

Pore size distribution(PSD)analysis of the supported CMSM. They were performed by permoporometry using the equipment and methodology reported by us [16]. Before the PSD measurement, the CMSM was dried under N₂ atmosphere at 3 bar and 300 °C for 24 h. He was used as carrier gas at 2 bar operational pressure and 70 °C for PSD measurements.

Carbon, hydrogen and oxygen elemental analysis. The chemical composition of the unsupported CMSMs were obtained using an automatic Micro analyzer (Thermo Scientific™ Flash smart elemental analyzer). Before analysis, the samples were dried at 300 °C under N₂ for 24 h to remove any adsorbed water molecules from the pores.

2.3. Pervaporation measurements

The pervaporation characterization of the CMSMs were carried out using the custom-made pervaporation setup reported by us [17] as represented by Fig. 1. Each CMSM was tested with 10 L of fresh feed at 37 °C (typical temperature used in fermentation studies) and permeate pressure of 12 mbar. Every hour the permeate was weighted and the composition determined by HPLC (Shimadzu, UFLC XR) equipped with a Refractive Index Detector (Waters 2414) and measuring column of Shimadzu, shim-pack (C18); each experiment was carried out for 6 h. For most of the cases, the average of the 6 samples measured at each hour is reported.

3. Results and discussion

PA, AA and water are the most relevant compounds in the separation of PA produced by bacterial fermentation, their relevant physical properties are listed in Table 1. Water is the smallest molecule with the higher dipole moment, PA is bigger than AA and PA has the smallest dipole-moment (more hydrophobic). The acids with one to four carbon atoms are completely miscible with water, their solubility decreases as the carbon chain length increases because the dipole forces become less important, and the dispersion forces become predominant (hydrophobic interaction). Monocarboxylic acids can form two strong H-bond in which carbonyl group acts as the hydrogen acceptor and hydroxyl group acts as the hydrogen donor which leads to the formation of self-associated intermolecular structures [18]. Due to hydrogen bonding, hydrophobic interactions and entropy, PA has more tendency to form dimers than AA [19], the hydrophilic groups are engaged in this interaction making the PA dimer more hydrophobic.

Considering only the molecular sieving mechanism of transport, water, the smallest molecule should preferentially permeate; however, if the pores are hydrophobic water will be the less permeable molecule. As adsorption diffusion mechanism of permeation is concerned, PA will interact more strongly with hydrophobic pores passing more preferentially than the more hydrophilic ones blocking at the same time the passage of the other smallest molecules. Considering the sizes of AA and PA, for PA separation, pores of around 0.5–0.6 nm are required.

Polyvinyl butyral (PVB) is a polymer containing polar and non-polar chemical domains (Scheme S1) enabling the formation of flexible green films with high tensile strength and higher elongation behavior [21]. PVB has oxygen groups that making it more reactive to decomposition at high temperatures. In this work, PVB is added to the dipping solution as pore forming agent, which, after carbonization, will leave in their place empty spaces (pores) and some carbon in the CMSM and as plasticizer for the phenolic polymer.

For the preparation of the CMSM, first an oligomer of resorcinol

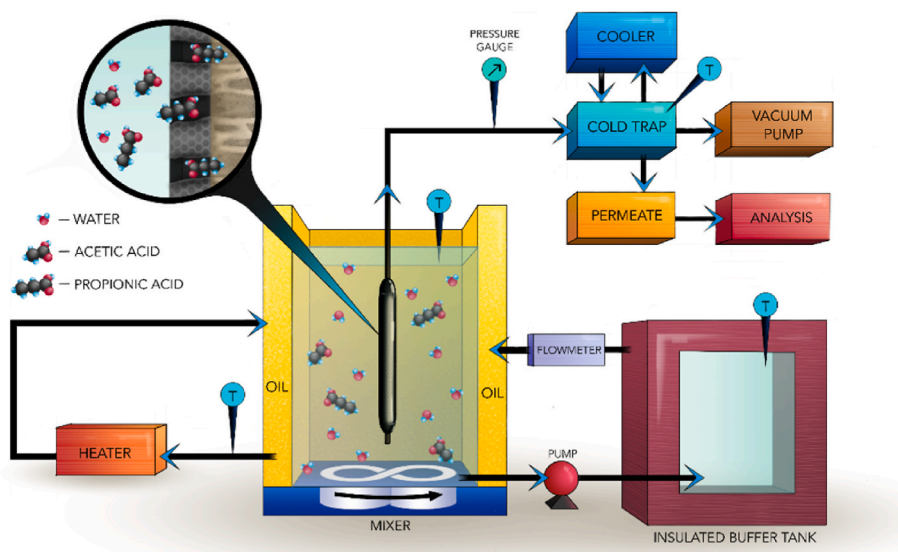


Fig. 1. The custom-built setup for pervaporation experiments of PA separation from aqueous solutions.

Table 1
Relevant properties of water, acetic acid (AA) and propionic acid (PA).

	MW [g mol ⁻¹]	Kin Dia ^a [nm]	BP ^b [°C]	MP ^c [°C]	Dip Mom. ^d [D]
H ₂ O	18	0.265	100	0	1.85
CH ₃ -COOH (AA)	60	0.436	118–119	16–17	1.74
CH ₃ -CH ₂ -COOH (PA)	74	0.480*	141	–20.5	0.63

^a Kinetic diameter.

^b Boiling point.

^c Melting point.

^d Dipole moment [Debye].

* [20].

phenolic resin is synthesized by condensation of resorcinol with formaldehyde using KOH as catalyst. The supported membranes were prepared by dip-coating of a tubular porous support in a dipping solution (Fig. S1); to form a layer on the support and avoid the infiltration of the solution inside the pores, the viscosity of the solution needs to be increased, therefore, the oligomer in the dipping solution was polymerized. Before dip-coating, PVB was added to the solution. The carbonization protocol is illustrated in Fig. S2.

Recently, our group reported hydrophilic CMSM prepared from phenolic resins for selective extraction of water/gas [22,23]; it was demonstrated that the pore size, pore size distribution and hydrophilicity/hydrophobicity of these CMSM could be tuned by changing the temperature of carbonization and the addition of hydrophilic nanoparticles. FTIR analysis indicates that between 550 °C and 650 °C, there is a sharp decrease in the oxygen content and in the 650–700 °C range. NMR relaxations and PSD analysis have shown that an increase in carbonization temperature during the preparation of composite Al-CMSMs causes a) the densification of the structure forming smaller pores and b) removal of hydrophilic functional groups making the pores more hydrophobic.

In Table 2, CHO composition of the non-supported CMSMs determined by elemental analysis (C, H and O Obs) are listed. From these values, the lowest molecular weight (MW) of the formula in which the number of CHN atoms are positive integer (considering at least two significant figures) were selected and C, H and O composition recalculated (Calc). It can be noticed that the composition of the

Table 2
Composition observed (Obs), molecular weight and composition calculated (Calc) of non-supported CMSM.

	Elemental analysis (%)		Number atoms	Elemental analysis (%)		Number atoms
	Obs	Calc		Obs	Calc	
750 P0				850 P0		
C	89.2	89.2	94	95.1	95.1	91
H	3.2	3.2	40	0.7	0.7	8
O	7.6	7.6	6	4.2	4.2	3
—		MW	1265	—	MW	1148
750P05				850P05		
C	92.1	92.1	95	96.3	96.3	83
H	2.7	2.7	33	0.6	0.59	6
O	5.2	5.2	4	3.1	3.1	2
—		MW	1239	—	MW	1034
750P1				850P1		
C	93.3	93.3	98	97.4	97.4	113
H	1.6	1.59	20	0.3	0.29	4
O	5.1	5.1	4	2.3	2.3	2
		MW	1260		MW	1392

observed and calculated structure matches very well. For the membranes carbonized at 750 °C, their molecular weights are similar, the number of carbons slightly increases, and hydrogen decreases as the amount of the PVB added increases. The MW of the membranes carbonized at 850 °C are higher than those at 750 °C, this is an indication of formation of aromatic-aromatic link [23] between the carbon chains with the consequent reduction of the size of the pores. For the membranes without PVB and with 0.5 % of PVB, the molecular weight of the basic structural unit decreases with the temperature; however, when the content of PVB increases to 1 wt percent, the opposite is observed, suggesting that some PVB has not been removed completely increasing the carbon content in the 850P1 membrane.

The hydrophilicity of the membranes was estimated by measuring the weight gained by the dry membrane, after its storage in an environment saturated with water, and the ratios of carbon to oxygen (C/O) and carbon to hydrogen (C/H) atoms in the CMSM (Table 3). Increasing the carbonization temperature from 750 °C to 850 °C the capacity to adsorb water is drastically reduced indicating the membrane is more hydrophobic; it is also reflected in the C/O ratio (the presence of O

Table 3

Water adsorbed (wt %) and relation of the number of C/O and C/H atoms * for non-supported membranes carbonized at 750 and 850 °C containing various contents of PVB.

	750P0	750P0.5	750P1	850P0	850P0.5	850P1
Water adsorbed (%)	17	15	13	7	5	4
Ratio C/O	15.7	23.8	24.5	30.2	41.5	56.5
Ratio C/H	2.4	2.9	4.9	11.4	13.7	28.3

makes the membrane more hydrophilic). The water adsorbed decreases with the addition of PVB which is related to the C/O ratio, the highest is for 850P1. The ratio C/H is an indication of the graphitization of the carbon structure consequently the hydrophobicity.

The pore size distribution of the CMSMs were determined by perm-porimetry; in this work, He was used instead of N₂ as inert gas to remove the water molecules condensed in the pores. The kinetic diameter of He (0.260 nm) is smaller than N₂ (0.364 nm), thus smaller pores can be detected; the results are shown in Fig. 2 and Fig. S3. The CMSMs carbonized at 750 °C show no uniform pore size distribution and most of the pores are bigger than 0.5 nm; the addition of PVB shifts the peaks to smaller pores. CMSM carbonized at 850 °C have more uniform pore distribution with a mean peak at around 0.5 nm, the percentage of pores (porosity) increase with content of the pore forming PVB, the. at higher carbonization temperatures, there are more condensation reducing the size of the pore. The pores bigger than 1 nm disappear with the addition of PVB because the produced carbon from PVB decomposition, fills the pores, and reduces the size of those smaller than 0.5 nm. For 850P1, 85 % of the pores are between 0.5 and 0.6 nm; one objective of this work. At 850 °C, at higher PVB content, is expected that the size and distribution will change and the pore forming properties of PVB will be more evident, these membranes will have different permeation properties and could be used for the separation of bigger molecules.

All the pervaporation studies were carried out for 6 h taking samples of analysis each hour; The observed permeation is a combination of molecular sieving (pore size and pore size distribution) and adsorption diffusion (interaction with the pore). Fig. 3a shows the pervaporation of only water; the water flux is almost constant with the time in all the cases. The membranes carbonized at 750 °C have higher water flux than those at 850 °C which is expected for a more hydrophilic membrane and the presence of bigger pores (Fig. 2); the water flux increases with the addition of PVB due to the increase of the porosity and change of the pore size distribution produced by the decomposition of the polymer. For the membranes carbonized at 850 °C, the number of pores in the 0.5 nm increases with the PVB content, increasing the water permeation.

Pervaporation studies of water solutions containing only PA or AA at 5 % wt% respectively are shown in Table S1 and Fig. 3. For PA, the highest purity is obtained by 850P1 (73.9 %) due to their higher

hydrophobicity and pore size distribution, in the range of 0.5–0.6 nm. For the same membrane, the AA content is 29.0 % mainly due to the higher dipolar moment than PA (1.74 and 0.63 D respectively); for the same reasons, the purity for PA and AA are lower for 750P1 than 850 P1. The purity of PA and AA increases with the addition of PVB, due to the increase in their fluxes since water flux is not dependent on the amount of PVB; but the flux increase considerably going from 0.5 to 1 % of PVB, which is attributed to the increase of porosity by the free space leave by PVB during carbonization.

Permeation studies were also carried out with samples containing in the same aqueous solution 5 % of PA and 5 % of AA (Fig. 3d) and Table 4, the results shows that the membranes are selective to PA-Considering the CMSM with the same composition of PVB in the dipping solution, PA selectivity is higher for those carbonized at 850 °C. Comparing PA purity in the aqueous solutions of 850P1, the sample containing both PA and AA has higher purity than that with only PA (88.3 and 73.9 % respectively); since the permeation mechanism of PA is adsorption diffusion, PA is adsorbed in the pores blocking the passage of the more hydrophilic and smaller AA. The PA flux is lower with 750P1, since it is less hydrophobic than 850 P1.

The pervaporation results of a solution containing the double of PA (10%PA+5%AA) are shown in Table 5 and Fig. 3e. The PA purity is like (5%PA+5%AA); but for AA, the lowest is for 750P1. The higher the concentration of AA, more is the blockage of the hydrophobic pores. For the solution with double concentration of AA (5%PA+10%AA) (Fig. 3f), the PA purity for 850P1 is lower (61.9 %) than the other cases where AA is present at more PA/AA ratio (Tables 4 and 5) due to the higher competition of AA with PA for the pores, the purity increases almost linearly with the PVB where the porosity and the hydrophobicity are larger.

4. Conclusions

In the fermentative production of PA, PA is an inhibitor for the reaction, therefore a CMSM able to separate PA from an aqueous mixture with AA (the main side-product) is required. To inhibit the permeation of water, the membrane should be hydrophobic. The kinetic diameter of PA and AA are close (0.480 and 0.436 nm respectively), but the main difference between them is their dipole moment (0.63 and 1.74 D respectively) and that PA has more tendency to form dimers by H-bond formation between two carboxylic groups making the dimer more hydrophobic.

Supported CMSM prepared by dipping into a solution containing resorcinol phenolic resin and PVB as pore forming and carbon source and carbonized at 750 and 850 °C. The C/O and C/H content in the membrane and water permeation studies shows that hydrophobicity, and PA selectivity of the membrane increase with the carbonization temperature and the content of PVB in the dipping solution.

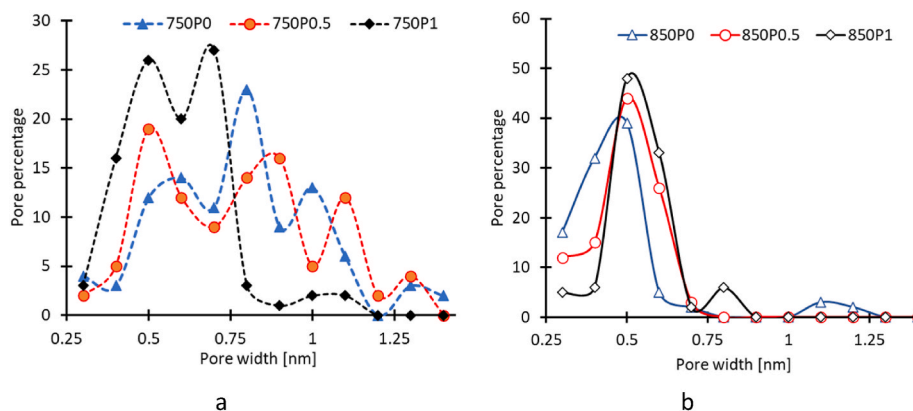


Fig. 2. Pore size distribution of CMSM carbonized at a) 750 °C and b) 850 °C.

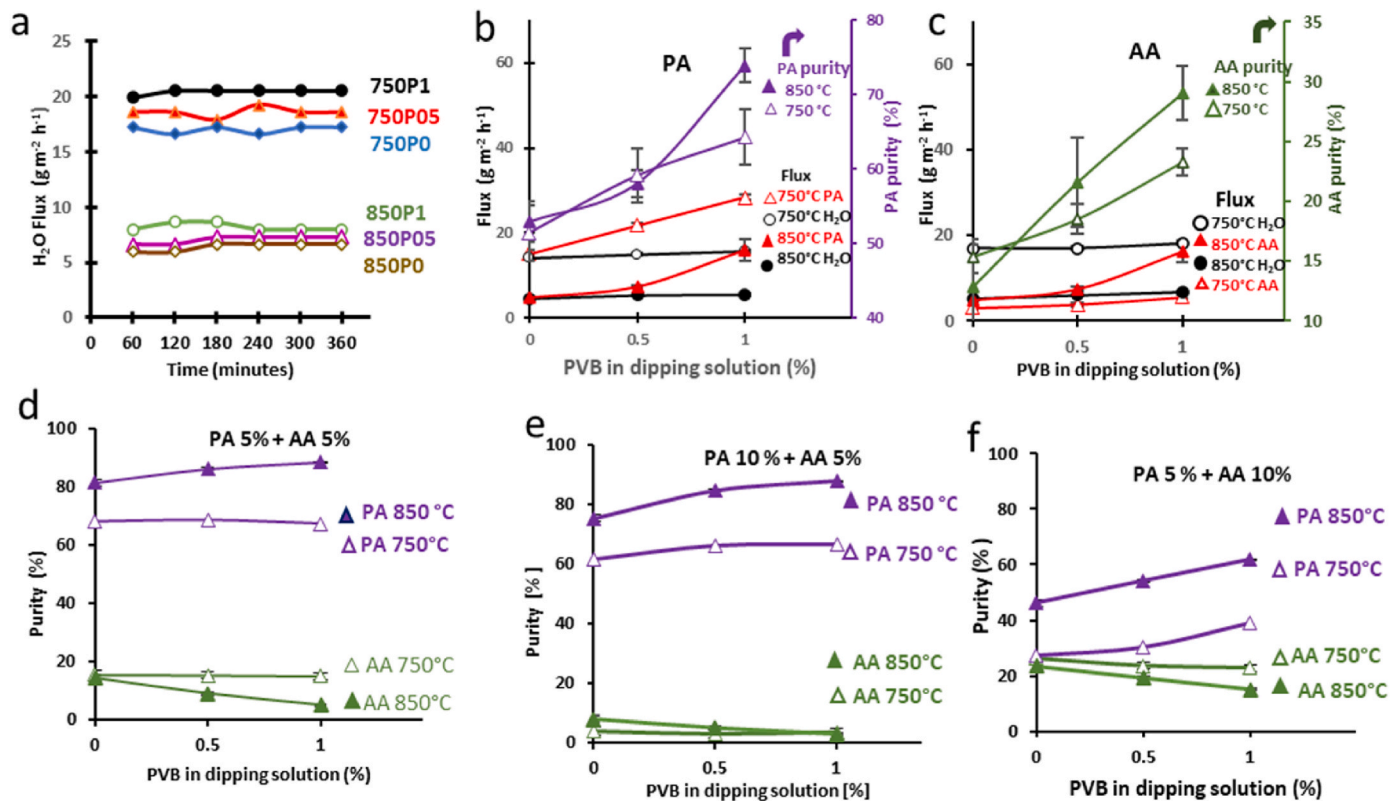


Fig. 3. Pervaporation studies of aqueous solutions of propionic acid (PA) and acetic acid (AA) using CMSMs carbonized at two temperatures (750 and 850 °C) from a dipping solution containing various concentration of PVB (0, 0.5 and 1 %) during the preparation of the membrane. The flux and the purity are referred to the permeated solution. (a) H₂O flux with the time for pervaporation of pure water. (b) PA flux and purity from a solution containing 5 % of PA. (c) AA flux and purity from a solution containing 5 % of AA. (d) PA and AA purity from a solution containing 5 % of PA and 5 % of AA. (e) PA and AA purity from a solution containing 10 % of PA and 5 % of AA. (f) PA and AA purity from a solution containing 5 % of PA and 10 % of AA.

Table 4

Effect of carbonization temperature and introduction of PVB in CMSMs on pervaporation properties of water solutions containing mixtures of 5 % of propionic acid (PA) and 5 % of acetic acid (AA).

	PA 5 %+ AA 5 %				
	Flux [g m ⁻² h ⁻¹]			Purity [%]	
	H ₂ O	PA	AA	PA	AA
750P	9.7	40.2	9.2	68.0	15.5
750P05	10.4	44.9	10.0	68.7	15.4
750P1	12.3	46.5	10.5	67.1	15.1
850P0	2.8	52.3	9.3	81.3	14.4
850P05	3.4	60.6	6.4	86.1	9.1
850P1	5.0	67.1	3.9	88.3	5.1

Permporometry studies revealed that the CMSMs carbonized at 850 °C have most of the pores in the range of 0.5–0.6 nm.

Comparing PA purity in the aqueous solutions of 850P1, the sample containing both PA and AA has higher purity than that with only PA (88.3 and 73.9 % respectively); since the permeation mechanism of PA is adsorption diffusion, PA is adsorbed in the pores blocking the passage of the more hydrophilic and smaller AA. PA with 88 % of purity was obtained by pervaporation of an aqueous solution containing 5 % of PA and 5 % of AA using a CMSM carbonized at 850 °C containing 1 % of PVB in the dipping solution.

The pore size, pore size distribution and hydrophilicity of the CMSM depend on the proportion of PVA used and the temperature of carbonization.

CRedit authorship contribution statement

Arash Rahimalimamaghani: Writing – original draft, Investigation,

Table 5

Effect of carbonization temperature and introduction of PVB in CMSMs on pervaporation properties of water solutions containing mixtures of 10 % of propionic acid (PA) and 5 % of acetic acid (AA), and 5 % of propionic acid (PA) and 10 % of acetic acid (AA).

	PA 10 % + AA 5 %					PA 5 % + AA 10 %				
	Flux [g m ⁻² h ⁻¹]			Purity [%]		Flux [g m ⁻² h ⁻¹]			Purity [%]	
	H ₂ O	PA	AA	PA	AA	H ₂ O	PA	AA	PA	AA
750P	7.0	12.3	0.8	61.4	3.8	8.3	5.0	4.8	27.6	26.4
750P05	7.0	14.9	0.7	66.2	2.9	10.3	6.9	5.3	30.5	23.7
750P1	8.3	18.6	1.0	66.7	3.6	11.1	11.4	6.6	39.2	22.8
850P0	1.4	6.4	0.7	75.3	7.8	2.1	3.2	1.7	46.2	23.7
850P05	1.4	11.5	0.7	84.6	4.9	2.5	5.2	1.9	54.2	19.4
850P1	2.2	20.7	0.7	87.8	2.8	3.6	9.8	2.4	61.9	15.2

Formal analysis, Conceptualization. **David A. Pacheco Tanaka:** Writing – original draft, Resources, Conceptualization. **Margot A. Llosa Tanco:** Writing – original draft, Validation, Formal analysis. **Fausto Gallucci:** Writing – review & editing, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.memsci.2024.123316>.

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